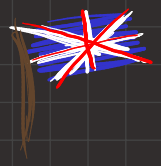
DISTRIBUITED SYSTEMS NOTES



Carlo Da Rome NOV. 5. 2026

CH. 1 G.P.E.

QUESTION 1: WHAT IS A DISTRIBUITED SYSTEM?

ANSWERE: A DISTRIBUITED SYSTEM IS A SET OF COMPUTERS THAT COOPERATE TO EXECUTE OPERATIONS OR TO STORE DATAS, THEY DON'T USE CLOCK FOR SINCRONIZATION, THE SINCRONIZATION IS POSSIBLE THROUGH MESSAGES.

- A LOT OF PROBLEMS IN THIS TYPE OF SYSTEM CAN BE TRACED BACK TO G.P.E. (global predicate evaluation) THAT IS EVALUATE IF THE SYSTEM SATISFIES A CONDITION.
- AN ASINCRONOS DYSTRIBUITED SYSTEM (A.D.S.) IS A SYSTEM WITH THE FOLLOWING PROPERTIES:

1. THE M PROCESSES IN THE SYSTEM IS REPRESENTED WITH THE SET $(P_1, \dots, P_m)$.
2. THE COMMUNICATION CHANNELS IS BASICALLY NOT FIFO, THIS MEANS THAT THE ORDER OF SENDED MESSAGES COULD BE DIFFERENT TO THE RECIEVE ORDER.
3. PROCESSES ARE STRONG CONNECTED, THIS MEANS THAT PROCESSES ARE ALL CONNECTED EACH OTHER.
4. CHANNELS ARE RELIABLE... WE CAN ASSUME THAT WE CANNOT HAVE DATA LOST OR COMPROMIZED.

5. SINCRONIZATION PASSES ONLY THROUGH MESSAGES

THIS IS A THEORETICAL MODEL, IT DOEST EXIST IN REALITY 'CAUSE IS NOT POSSIBLE TO REALIZE THAT BUT IT IS A GOOD ABSTRACTION BECAUSE.

1. IS VERY SIMPLE AND DOESN'T HAVE ANY BOUND IN TERM OF SPEED OR RETARD, IF ONE SOLUTION IS GOOD ON THIS MODEL IT IS ALSO GOOD IN REAL LIFE.

2. SOME LAYERS IN REAL SYSTEMS FOLLOW THIS MODEL (RULE 4 IS IMPOSSIBLE TO SATISFIES IN REAL LIFE)

- IN A DISTRIBUITED COMPUTATION WE CAN HAVE 3 TYPE OF EVENTS

1. INTERNAL EVENT.

IT ONLY CHANGE THE STATE OF THE PROCESS THAT EXECUTES THE EVENT.

2. EXTERNAL EVENT.

2.1 SEND(m)

2.2 RECIEVE(m)

THESE TWO TYPE OF EVENTS ATTACH A MESSAGE INDICATED WITH THE LEETER "m"

CH.2 GLOBAL STATES, CUT AND CONSISTENCY

QUESTION 2: WHAT IS CONSISTENCY?

ANSWERE: IN A D.S. WE HAVE CONSISTENCY WHEN THE MONITOR PROCESS p_0 (PROCESS THAT MAKE G.P.C.) RECIEVE COMPATIBLE STATES.

- HISTORY.

IN DISTRIBUITED SYSTEM WE HAVE TWO TYPE OF

HISTORY...

1. LOCAL HISTORY.

IS THE SET OF THE EVENTS EXECUTED BY A SPECIFIC PROCESS P_i

$h_i = e_i^1 e_i^2 e_i^3 \dots$ POTENCIALLY INFINITE...

↑ WHERE i IS REFERRED TO THE PROCESS P_i

2. GLOBAL HISTORY.

IS THE SET OF ALL THE EVENTS EXECUTED BY ALL THE PROCESSES IN THE DISTRIBUTED SYSTEM.

$H = h_1 \cup h_2 \cup h_3 \cup \dots \cup h_m$

$H = e_1^1 \dots e_1^{\dots} \cup e_2^1 \dots e_2^{\dots} \cup [\dots]$

- CAUSAL PRECEDENCE.

THE CAUSAL PRECEDENCE "$\rightarrow$" IS A RELATION ON THE EVENTS IN H

IF $e \rightarrow e' \Rightarrow e$ INFLUENCE e' AND FOLLOWS THE FOLLOWING PROPERTIES:

1. IF e, e' IS EVENT OF THE SAME PROCESS P_i AND e PRECEDES e' IN THE LOCAL HISTORY h_i THEN $e \rightarrow e'$.
2. IF e = SEND(m) AND e' = RECIEVE(m) THEN $e \rightarrow e'$
3. IF $e \rightarrow e'$ AND $e' \rightarrow e''$ THEN $e \rightarrow e''$.

- GLOBAL STATES, CUT, RUN

- TO PERFORM A G.P.E. THE MONITOR p_0 REQUIRES THE GLOBAL STATE OF THE SISTEM... THE GLOBAL STATE Σ IS DEFINED AS $\Sigma = \sigma_1, \sigma_2, \dots, \sigma_m$ WHERE σ_i IS THE LOCAL STATE OF P_i.
- A CUT IS A SUBSET IN THE GLOBAL HISTORY.

$C = h_1^{e_1} \cup \dots h_m^{e_m}$.

THE TUPLE $(e_1,\dots,e_m)$ INDICATE THE LAST EVENT TO CONSIDER IN THE LOCAL HISTORY.

- A RUN IS A TOTAL ORDER ON H, IF e PRECEDES e' IN LOCAL HISTORY h_i OF p_i THEN e PRECEDES e' IN R.
- A CUT IS CONSISTENT IF LEFT CLOSED OR RATHER: IF $\forall e, e' \in H$ AND $e' \in C$ AND $e \to e' \Rightarrow e \in C$.
- A RUN IS CONSISTENT IF $\forall e, e'$ $e \to e' \Rightarrow$ e APPEARS BEFORE e' IN R.

CH. 3 MONITORING A D.S.

TO MONITOR A D.S. WE CAN APPLY TWO TYPE OF PREDICATE... ACTIVE MONITOR OR PASSIVE MONITOR.

ACTIVE MONITOR.

IN SYSTEMS WITH ACTIVE MONITOR, THE MONITOR PROCESS p_0 ASKS CONTINOUSLY TO THE OTHER PROCESSES THEIR LOCAL STATE. WE CAN NOTICE SOME PROBLEMS IN THIS TYPE OF SYSTEM.

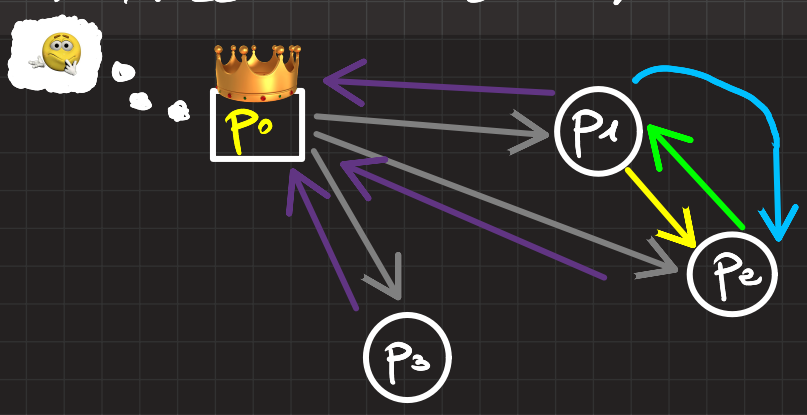
1. GLOBAL STATES THAT THE MONITOR RETRIEVE CAN BE INCONCOSISTENT AND INCREASING THE FREQUENCY OF THE REQUESTS THE PROBLEM IS NOT FIXED.

2. GHOST DEADLOCK

WE ALSO COULD HAVE GHOST DEADLOCK...

BASIC EXAMPLE OF GHOST DEADLOCK.

EXAMPLE 1.



IN THE EXAMPLE WHE HAVE THE FOLLOWING SITUATION:

P_1 SEND TO P_2 A MESSAGE AND IT WAIT FOR A CONFIRMATION MESSAGE FROM P_2

P_2 SEND TO P_1 THE CONFIRMATION MESSAGE ... AND WAIT FOR A CONFIRMATION MESSAGE FROM P_1 ...

IF THE MONITOR P_0 SENDS THE STATE REQUEST IT CAN ARRIVES TO P_1 AND P_2 IN THE FOLLOWING MOMENT...

P_1 : WAITING FOR P_2 CONFIRMATION MESSAGE
P_2 : WAITING FOR P_1 CONFIRMATION MESSAGE

(THIS IS CAUSED BY THE IMPREVEDIBLE RETARDS ON THE COMMUNICATION LINE)

THE GLOBAL STATE RETRIVED IS AN INCONSISTENT STATE AND IT CONTAINS A GHOST DEADLOCK, OR RATHER A DEADLOCK THAT DOESN'T EXIST IN THE SYSTEM BUT EXIST IN G.P.E..

- PASSIVE MONITOR.

IN SYSTEM WITH PASSIVE MONITOR THE MONITOR PROCESS P_0 DOESN'T ASK TO THE OTHER THEIR STATE BUT THE PROCESSES IN D.S. SENDS TO THE MONITOR ONLY RELEVANT INFORMATIONS.